

The face lattice of the irreducible-Ferrers polytope is a product of triangles

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Abstract

Beeloo-Sauerbier Couvée and Neri associated, for every integer $d \geq 3$, an integral polytope $\mathfrak{P}_d$ with irreducible Ferrers diagrams and conjectured that $\mathfrak{P}_d$ is combinatorially equivalent to a Cartesian product of $d-2$ triangles. Their construction includes a linearly isomorphic nonnegative affine slice $\mathfrak{L}_d$ and a complete description of its vertices: after grouping the coordinates into $d-2$ triples, each vertex has exactly one positive coordinate in every triple, and all 3^{d-2} possible support patterns occur. We show that this vertex-support description already determines the whole face lattice. More generally, a polytope obtained by intersecting an affine subspace with a nonnegative orthant, whose vertices realize exactly the one-per-block supports, has the face lattice of a product of simplices. Applying this observation proves the product-of-triangles conjecture for every $d \geq 3$. It also gives the complete face enumerator, simplicity, and the number of facets of $\mathfrak{P}_d$.

1 Introduction

Ferrers-diagram rank-metric codes connect algebraic coding theory with the combinatorics of partitions and convex polytopes. In recent work, Beeloo-Sauerbier Couvée and Neri [1] introduced a reduction theory for Ferrers diagrams in the Etzion–Silberstein conjecture. For every distance parameter $d \geq 3$, they encoded the irreducible diagram pairs as the integer points of an integral polytope $\mathfrak{P}_d$.

The same paper computed the face vectors of $\mathfrak{P}_d$ for $3 \leq d \leq 7$. These vectors agree with those of the Cartesian product of $d-2$ triangles, leading to the following conjecture [1, Conjecture 5.28]:

For every $d \geq 3$, the polytope $\mathfrak{P}_d$ is combinatorially equivalent to the Cartesian product of $d-2$ triangles.

The purpose of this note is to prove that conjecture. The argument uses no additional enumeration. The key input is the vertex theorem already proved in [1, Proposition 5.25]. After a sequence of linear isomorphisms, the polytope is presented as a nonnegative affine slice with coordinates grouped into triples. Its vertices realize all possible choices of one positive coordinate per triple. In such a presentation, a face is determined by the coordinates that

vanish on its relative interior, and averaging compatible vertices realizes every allowable vanishing pattern. Those patterns are exactly the faces of a product of triangles.

We isolate this argument as a general block-support lemma for products of simplices. The Ferrers-polytope conjecture is its immediate three-coordinate specialization. This formulation separates the new observation from the substantial prior work in [1]: the definition and integrality of $\mathfrak{P}_d$, the affine model $\mathfrak{L}_d$, and the complete vertex classification all belong to Beeloo-Sauerbier Couvée and Neri. The new result is the recovery of the full face lattice from that classification.

Throughout, a *polytope* is a nonempty bounded polyhedron. Two polytopes are *combinatorially equivalent* if their face lattices, including the empty face, are isomorphic. We write Δ_{r-1} for an $(r-1)$ -simplex.

2 Faces of nonnegative affine slices

We first recall directly, in the form needed below, how faces are encoded in a nonnegative affine-slice presentation.

Let $A \subseteq \mathbb{R}^N$ be an affine subspace and let

$$P = A \cap \mathbb{R}_{\geq 0}^N$$

be a polytope. For a nonempty face F of P , choose $p \in \text{relint}(F)$ and define

$$Z(F) = \{j \in [N] : p_j = 0\}.$$

The set $Z(F)$ is independent of the chosen relative-interior point, and

$$F = P \cap \bigcap_{j \in Z(F)} \{x_j = 0\}. \quad (2.1)$$

For completeness, let y belong to the right-hand side of (2.1). Every coordinate of p outside $Z(F)$ is positive, so for sufficiently small $\varepsilon > 0$,

$$w = (1 + \varepsilon)p - \varepsilon y$$

still belongs to $A \cap \mathbb{R}_{\geq 0}^N = P$. Since $p = (w + \varepsilon y)/(1 + \varepsilon)$, the face property and $p \in \text{relint}(F)$ imply $w, y \in F$. Thus (2.1) holds. Conversely, an intersection with coordinate hyperplanes as in (2.1), when nonempty, is a face of P .

We now group the coordinates. Let I be a finite index set and, for every $i \in I$, let E_i be a finite set with $|E_i| \geq 2$. The ambient coordinates are indexed by pairs (i, e) with $i \in I$ and $e \in E_i$.

Lemma 2.1 (complete block-support lemma). *Let*

$$P = A \cap \mathbb{R}_{\geq 0}^{\sum_{i \in I} |E_i|}$$

be a polytope. Suppose its vertices are precisely

$$v_\sigma, \quad \sigma \in \prod_{i \in I} E_i,$$

and their supports satisfy

$$(v_\sigma)_{i,e} > 0 \iff e = \sigma_i. \quad (2.2)$$

Then the face lattice of P is isomorphic to the face lattice of

$$\prod_{i \in I} \Delta_{|E_i|-1}.$$

Proof. Let F be a nonempty face of P . For every $i \in I$, its zero set $Z(F)$ cannot contain all coordinates (i, e) with $e \in E_i$: otherwise every vertex of F would have zero support in the entire i -th block, contradicting (2.2). Hence

$$S_i(F) = \{e \in E_i : (i, e) \notin Z(F)\}$$

is nonempty. Equation (2.1) shows that F is uniquely determined by the tuple $(S_i(F))_{i \in I}$.

It remains to show that every tuple of nonempty subsets occurs. Fix nonempty $S_i \subseteq E_i$ and set

$$\Sigma_S = \{\sigma \in \prod_{i \in I} E_i : \sigma_i \in S_i \text{ for every } i\}.$$

This set is nonempty. Let

$$q_S = \frac{1}{|\Sigma_S|} \sum_{\sigma \in \Sigma_S} v_\sigma.$$

By (2.2), the coordinate (i, e) of q_S is positive exactly when $e \in S_i$. Therefore the zero set of q_S is precisely

$$Z_S = \{(i, e) : e \notin S_i\}.$$

The coordinate face

$$F_S = P \cap \bigcap_{(i,e) \in Z_S} \{x_{i,e} = 0\}$$

is nonempty, contains q_S , and has no coordinate inequality beyond those in Z_S active at q_S . Thus $q_S \in \text{relint}(F_S)$ and $Z(F_S) = Z_S$. In particular, distinct tuples S give distinct faces.

We have obtained a bijection between the nonempty faces of P and tuples of nonempty subsets $S_i \subseteq E_i$. Under this bijection, face containment is coordinatewise inclusion of the subsets. This is exactly the nonempty face poset of $\prod_i \Delta_{|E_i|-1}$. Matching the unique empty faces completes the face-lattice isomorphism. $\square$

Remark 2.2. The hypotheses require the complete and exact support description (2.2). A vertex count alone does not determine the face lattice. Strict positivity of the selected coordinate is also essential: without it, averaging compatible vertices need not realize the intended zero pattern.

The lemma proves only combinatorial equivalence. It does not produce an affine or a unimodular equivalence between P and the product of simplices.

3 The Ferrers polytopes

We now apply Lemma 2.1 to the polytopes of [1]. Fix $d \geq 3$ and put

$$k = d - 2.$$

Beeloo-Sauerbier Couvée and Neri construct four linearly isomorphic polytopes

$$\mathfrak{P}_d, \quad \mathfrak{P}'_d, \quad \mathfrak{K}_d, \quad \mathfrak{L}_d. \quad (3.1)$$

The first transformation is unimodular [1, Lemma 5.23]. The latter two arise from injective linear maps whose restrictions are linear isomorphisms onto their images [1, Lemma 5.24 and the definitions following it]. Consequently, all four polytopes have isomorphic face lattices.

The model relevant here is $\mathfrak{L}_d \subseteq \mathbb{R}^{3k}$. Write its coordinates as

$$(x_1, \dots, x_k, y_1, \dots, y_k, z_1, \dots, z_k).$$

Equation (11) of [1] presents $\mathfrak{L}_d$ as the intersection of the nonnegative orthant with k affine equations. The precise coefficient matrix of those equations will not be needed here; what matters is that the only inequalities in this presentation are

$$x_i \geq 0, \quad y_i \geq 0, \quad z_i \geq 0 \quad (i \in [k]). \quad (3.2)$$

Proposition 5.25 of [1] gives the complete vertex set. For every word

$$\sigma \in \{X, Y, Z\}^k,$$

there is exactly one vertex v_σ , and in the i -th coordinate triple

$$(x_i, y_i, z_i)$$

the coordinate selected by σ_i is strictly positive while the other two coordinates are zero. Conversely, every vertex is obtained in this way. The strict positivity follows in [1] from the explicit consecutive-block formula in the proof of Proposition 5.25.

We can therefore take $I = [k]$ and $E_i = \{X, Y, Z\}$ in Lemma 2.1.

Theorem 3.1. *For every integer $d \geq 3$,*

$$\mathfrak{P}_d \simeq_{\text{comb}} (\Delta_2)^{d-2}. \quad (3.3)$$

In particular, Conjecture 5.28 of [1] holds for all $d \geq 3$.

Proof. The affine-slice presentation (3.2) and the vertex-support theorem just described satisfy every hypothesis of Lemma 2.1. Hence

$$\mathfrak{L}_d \simeq_{\text{comb}} (\Delta_2)^k.$$

The linear isomorphisms in (3.1) preserve face lattices. Substituting $k = d - 2$ proves (3.3). $\square$

Corollary 3.2. *Let $k = d - 2$. The polytope $\mathfrak{P}_d$ is a simple polytope of dimension $2k$, with $3k$ facets and 3^k vertices. Every vertex has degree $2k$.*

Proof. These properties hold for a product of k two-dimensional simplices and are preserved by combinatorial equivalence. The vertex count also agrees with [1, Theorem 5.20]. $\square$

Corollary 3.3. *If $f_j(\mathfrak{P}_d)$ denotes the number of j -dimensional faces, then*

$$\sum_{j=0}^{2k} f_j(\mathfrak{P}_d) t^j = (3 + 3t + t^2)^k. \quad (3.4)$$

Equivalently,

$$f_j(\mathfrak{P}_d) = \sum_{b=0}^{\lfloor j/2 \rfloor} \frac{k!}{(j-2b)! b! (k-j+b)!} 3^{k-b}, \quad (3.5)$$

where a summand is understood to vanish when one of the factorial arguments is negative.

Proof. The face polynomial of a triangle, with the empty face omitted, is $3 + 3t + t^2$. Face dimensions add under Cartesian products, giving (3.4). To obtain (3.5), choose b factors contributing a two-dimensional face, $j - 2b$ factors contributing an edge, and all remaining factors contributing a vertex. The edge and vertex choices contribute $3^{j-2b} 3^{k-j+b} = 3^{k-b}$. $\square$

Remark 3.4. Formula (3.4) explains, for every d , the face vectors computed in [1] for $3 \leq d \leq 7$. These enumerative statements are consequences of the single face-lattice theorem, not separate independent results.

4 Scope and relation to prior work

Theorem 3.1 is short because the difficult structural input was already available. The construction of $\mathfrak{P}_d$, its interpretation in terms of irreducible Ferrers diagrams, its integrality, and the exact 3^{d-2} -vertex classification are all results of [1]. The present note adds the observation that their support classification, together with the nonnegative affine-slice presentation, determines every face rather than only the vertices.

The result does not settle the Etzion–Silberstein conjecture itself. The polytopes $\mathfrak{P}_d$ parametrize the irreducible cases identified in [1], but knowing their face lattices does not construct maximum Ferrers diagram codes at their integer points. Nor does Theorem 3.1 assert that $\mathfrak{P}_d$ is affinely or unimodularly equivalent to a product of triangles. Such stronger equivalences would require metric or lattice data not encoded by the face lattice.

As of 24 August 2026, the arXiv record of [1] contains only version v1 and the authors’ public repository contains only its initial commit. Exact searches for Conjecture 5.28 and its product-of-triangles formulation found no direct public proof. This search boundary supports the relevance of the result but is not a claim of exhaustive global priority.

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References

- [1] H. Beeloo-Sauerbier Couvée and A. Neri, *Irreducible Ferrers diagrams in the Etzion–Silberstein conjecture*, arXiv:2604.27868v1 (2026). [arXiv:2604.27868](https://arxiv.org/abs/2604.27868).